

DBMS Unit 4 — Transactions & Concurrency Control Complete Notes

Database Management Systems (DBMS) · Unit 4 · Transactions & Concurrency Control · 2024

Overview

These notes cover **Transactions & Concurrency Control**. Design, query and manage relational databases — from ER modelling and SQL to normalization, transactions and recovery. Each section states the core idea, a worked intuition and the points examiners look for.

1. ACID Properties

Atomicity, consistency, isolation and durability define correct transaction behaviour.

Why it matters: ACID Properties appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of acid properties in one line.
- A small example showing acid properties in practice.
- Common mistakes students make and how to avoid them.

2. Serializability

A schedule is conflict serializable if its precedence graph is acyclic, making it equivalent to some serial order.

Why it matters: Serializability appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of serializability in one line.
- A small example showing serializability in practice.
- Common mistakes students make and how to avoid them.

3. Two-Phase Locking

2PL has a growing and a shrinking phase; strict 2PL holds exclusive locks until commit to avoid cascading aborts.

Why it matters: Two-Phase Locking appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of two-phase locking in one line.
- A small example showing two-phase locking in practice.
- Common mistakes students make and how to avoid them.

4. Timestamp Ordering

Timestamp protocols order conflicting operations by transaction timestamps, with Thomas' write rule skipping obsolete writes.

Why it matters: Timestamp Ordering appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of timestamp ordering in one line.
- A small example showing timestamp ordering in practice.
- Common mistakes students make and how to avoid them.

5. Deadlock Handling in Databases

Wait-die and wound-wait schemes prevent deadlocks, while wait-for graphs detect them.

Why it matters: Deadlock Handling in Databases appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of deadlock handling in databases in one line.
- A small example showing deadlock handling in databases in practice.
- Common mistakes students make and how to avoid them.

6. Isolation Levels

Read uncommitted, read committed, repeatable read and serializable trade consistency for concurrency.

Why it matters: Isolation Levels appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of isolation levels in one line.
- A small example showing isolation levels in practice.
- Common mistakes students make and how to avoid them.

Quick revision

ACID Properties · Serializability · Two-Phase Locking · Timestamp Ordering · Deadlock Handling in Databases · Isolation Levels